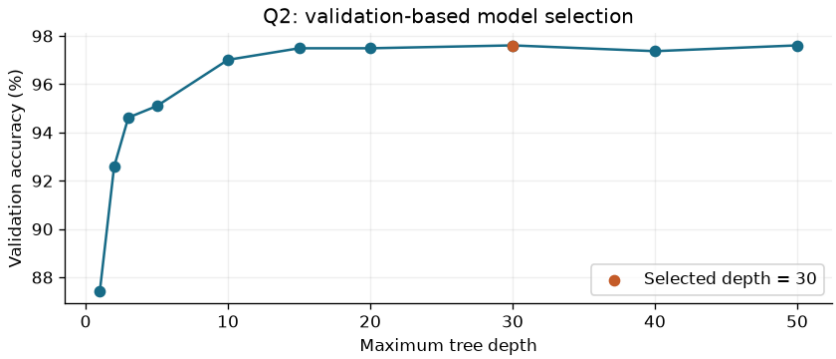
# Q2: SMS decision tree

The 5,574 raw messages are split into 3,901 training, 836 validation and 837 test rows (70/15/15, rounded). Both splits are stratified with random_state=42. CountVectorizer is fitted only on training messages, producing 7,179 word features. Validation and test messages use transform only.

A manual loop tests ten max_depth values. Each DecisionTreeClassifier uses the default Gini criterion and random_state=42. The highest validation score is selected; ties choose the smaller depth.

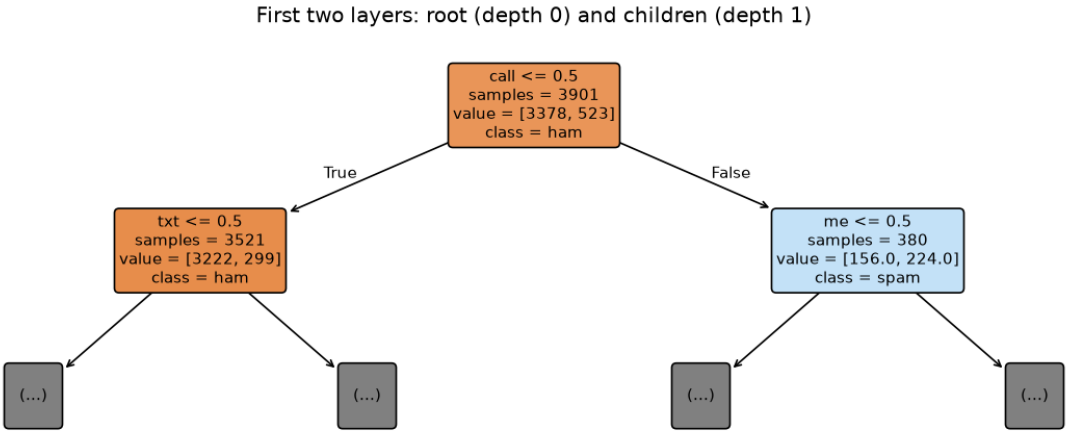
## Validation results

```
max_depth  validation_accuracy
        1  87.440191%
        2  92.583732%
        3  94.617225%
        5  95.095694%
       10  97.009569%
       15  97.488038%
       20  97.488038%
       30  97.607656%
       40  97.368421%
       50  97.607656%
Best max_depth: 30
Validation: 97.607656%
```




Depths 30 and 50 tie at **97.6077%**. Depth **30** is selected. The selected training-fitted model is tested once without refitting: **96.5352% accuracy (808/837 correct)**.

## First two layers of the selected tree




The diagram shows the root and its children; dots indicate deeper branches. The root checks whether "call" appears (count <= 0.5). This makes sense because promotional messages may ask the recipient to call a number. Legitimate messages also use the word, so later splits are needed.

Ham accounts for 86.6% of messages, so accuracy alone does not measure spam recall. The 403 duplicate rows beyond first occurrences are retained; repeated messages may cross splits and make results optimistic.

Data: supplied sms.csv (SMS Spam Collection).